Federated Learning for IoV Adaptive Vehicle Clusters: A Dynamic Gradient Compression Strategy

Ling Xing, Pengcheng Zhao, Jianping Gao, Honghai Wu, Huahong Ma, and Xiaohui Zhang

Abstract—**Federated learning (FL) has been extensively utilized in distributed learning scenarios for the Internet of Vehicles (IoV). However, two key challenges exist: 1) gradients are frequently transmitted between vehicles during FL training, which reduces the communication timeliness between the traffic participants and 2) the fixed gradient compression method cannot sufficiently adapt to the dynamic IoV network topology. Therefore, we design an adaptive vehicle clustering method constructed according to multiple attributes, such as computational resources, communication distance, and latency. Accordingly, we propose a dynamic gradient compression strategy that filters similar local training models between vehicles and uses the Wasserstein distance to compute a sparsity threshold. This threshold acts as a dynamic compression factor that compresses gradient model parameters, reducing redundant parameter transmission. Furthermore, we conduct experiments using two datasets to evaluate the proposed strategy's effectiveness. The compression ratio improved by 132- and 178-fold compared to the baselines, and the aggregated accuracy increased by an average of 10.13%. Additionally, experiments incorporating communication noise revealed that the aggregation model of the signal noise ratio is -29 dB.**

Index Terms—**Adaptive vehicle cluster, federated learning (FL), gradient compression, Internet of vehicles (IoV).**

I. Introduction

The Internet of Vehicles (IoV) network facilitates seamless connectivity between individuals, vehicles, roads, and cloud infrastructure. Each vehicle is an entity that has been adaptively linked, enabling information distribution [1], [2]. The IoV data communication structure includes vehicle-to-vehicle (V2V), -person (V2P), -roadside units (V2R), -sensors (V2S), and -infrastructure (V2I) cellular networks [3], [4]. In addition, the communication methods between vehicles and entities mainly utilize the dedicated short-range communication (DSRC) and long-term evolution-vehicle (LTE-V) system frameworks for the IoV [5], [6]. However, challenges, such as high communication overhead, privacy concerns during data transmission, broadcasting, and information sharing among vehicle clients, exist. These challenges arise due to factors, such as data source variability, heterogeneity, dynamic network topology, and autonomous connectivity [7], [8].

Federated learning (FL) is a distributed learning mechanism that can be applied to collaborative data security for IoV generalizability, inference, and modeling [9], [10], [11]. FL was proposed by Bonawitz et al. [12] and is used to achieve data privacy and security. It is particularly suitable for scenarios involving sensitive data, such as medical records, personal information recognition, and intelligent vehicle networking [13], [14], [15], [16]. Moreover, FL training data remains with the participant (i.e., the mobile client) [17], [18], and the updated machine learning parameters for local clients are transmitted to the central server, allowing for participation in secure aggregation [19]. Thus, the widespread application of FL is crucial for the efficient recognition of traffic signs, which improves the capabilities of autonomous driving technology [20], [21], [22]. FL also enhances the efficiency of vehicle–road collaborative training, interaction, and information sharing. Currently, dynamic clustering and asynchronous FL are used to provide basic support for the application and deployment of intelligent transportation systems [23]. Fig. 1 shows a wide range of application scenarios (e.g., distributed training and intelligent decision making in complex traffic scenarios) that combine the IoV with FL architecture. Vehicles and infrastructure (e.g., roadside units and traffic lights) within the IoV network are trained collaboratively using a multilevel FL framework. Thus, integrating FL and vehicle–road–cloud architecture allows vehicle-side sensors to collect perception data for local model training. Also, vehicle clients are used for local training while the roads and cloud network build a global aggregate model to inform decision making.

However, despite its advantages, FL still faces several issues. For instance, vehicle dynamics can lead to a decrease in model aggregation accuracy [24]. Large-scale FL distributed training utilizes a substantial communication bandwidth to exchange gradients, limiting the scalability of multinode training and resulting in an expensive network infrastructure. Additionally, FL often struggles to achieve satisfactory

Received 24 December 2024; revised 4 January 2025; accepted 27 January 2025. Date of publication 4 February 2025; date of current version 23 May 2025. This work was supported by the National Natural Science Foundation of China under Grant U22A2069, Grant U23A20272, and Grant 62272146. *(Corresponding author: Jianping Gao.)*

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Digital Object Identifier 10.1109/JIOT.2025.3538551

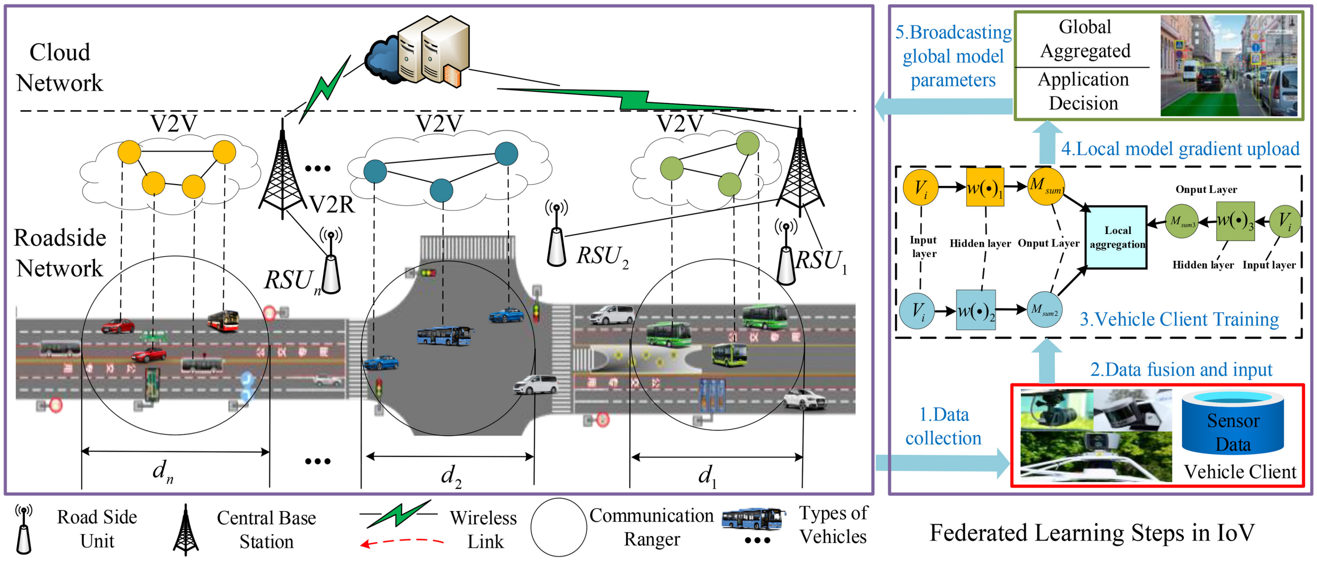

Fig. 1. Overview of the IoV scenario and the FL mechanism.

computational and time costs during model training, resulting in a communication bottleneck that becomes more prominent when dealing with complex models with multiple features [25]. To address these issues, current research focuses on improving communication efficiency and reducing computing costs in two ways: 1) local client training models can be compressed to reduce the overall parameter size of the data transmitted to the global aggregation and 2) the total number of training iterations can be reduced to decrease the number of gradient transmissions during training, reducing the overall amount of transmitted data.

Existing gradient compression methods that provide a trade-off between communication cost and training accuracy can be categorized into two categories: 1) gradient quantization and 2) sparsification [26]. Gradient quantization reduces the gradient's representation size by converting it into an integer with fewer bits, reducing communication and computational overhead. However, quantized gradients are prone to information loss, and adjusting the scale can be challenging. In contrast, the gradient sparsification method obtains a domain sparser than the original. Meanwhile, sparse updates directly affect model convergence. However, determining an appropriate sparse threshold in practical applications can be challenging. The deep gradient compression method uses momentum correction, gradient pruning, factor hiding, and warm-up training to reduce the number of interaction model parameters [27]. While these methods effectively sparsify FL gradient aggregation, they tend to be less robust when applied to dynamic network structures.

Based on the deep quantization approach, the model gradient can be compressed using the optimal loss-less encoding or local self-encoding method, which lacks robustness in the dynamic network topology [28], [29], [30]. In [31] and [32], the ternary sparse gradient compression method was used to reduce the distributed learning computational complexity, ensuring the calculation model's robustness. Similarly, in [33] and [34], the gradient was sparsely compressed in layers, and gradients were randomly selected for sharing and aggregation. Li et al. [35] proposed a fine-grained gradient compression method based on the model's layers to reduce the number of interactive iterations between the device and server. Likewise, Jiang et al. [36] proposed a key–value pair compression method that compresses the sparse gradient vector.

Gradient compression [37], recently proposed by researchers, is one of the most effective methods for top-K gradient selection, as it compresses the sent message size per node (0.1% gradient size). After the original gradient of each layer is selected, the algorithm accumulates the remaining gradient as a residual, which is subjected to a new round of iterations. This method achieves high accuracy with the same convergence rate.

However, these methods may not effectively optimize the available communication resources when dealing with dynamic vehicles in the IoV. Table I reviews the applications of gradient quantization and sparsification in FL, summarizing and comparing various research and technical features with gradient compression. According to the review, FL gradient sparsification is affected by the model gradient information loss or transmission delay. This is due to the frequent changes in node topology within dynamic networks, affecting the local aggregation model's integrity. Meanwhile, the sparsification strategy cannot adapt to the requirements of various nodes, leading to poor model robustness. This indicates that existing methods fail to guarantee a high global convergence rate and aggregation accuracy across various complex traffic scenarios. Specifically, in dynamic IoV scenarios, the inherent properties of sparse thresholds reduce model aggregation accuracy. To solve this problem, we propose a dynamic gradient compression FL strategy for adaptive vehicle IoV clustering. This article's main contributions are as follows.

1) We study the multidimensional attributes of vehicle network scenarios and clusters, such as communication distance, latency, and computing resources. Then, we propose a mechanism for building adaptive vehicle clusters.

TABLE I
RELATED WORK

References and years	Introduction	The target or related work
[38] --2017	Gradient Quantization	The compressed gradients for each node layer are transmitted during each iteration.
[39] --2017	Gradient Pruning Technique and Hierarchical Ternary	Gradients are quantized to ternary precision before being passed to the parameter server, reducing gradient communication.
[40] --2018	Async-Average-Gradient	The K-AVG algorithm utilizes the average gradient compression method and sets a larger step size to facilitate fast model convergence.
[41] --2018	Adacomp Compression Method	The maximum threshold updating method is selected in the model to avoid global aggregation information loss by considering the computational latency of real-time parameters.
[42] --2018	Local SGD Update Variant	By adjusting the number of model iterations in the communication interval, the decoupling of local delay and aggregation can be achieved, improving accuracy.
[43] --2019	Sparse Gradient Compression	The momentum approximation method is used to reduce the amount of memory transmitted and compensate for the long-term information loss caused by compression.
[44] --2019	Sparse Binary Compression	Combining binarization with optimal weight autoencoders, achieving gradient sparsity selection to facilitate compression.
[45] --2019	AdaComm Adaptive Communication	Optimizing the weight update rate of clients in distributed computing accelerates model convergence.
[46] --2020	Clustergrad Compressed Gradient	The K-means algorithm identifies the basic gradient far away from the origin, and clustering quantizes the gradient to complete compression.
[47] --2020	Local Gradient Compression and Spectrum Allocation	Selecting gradients and compression ratios in batch sizes based on local client atomic matrix sizes.
[48] --2021	Counting Sketch Gradient Compression	The wireless counting sketch algorithm is used to select gradients.
[49] --2021	Model Gradient Update Compression	Likelihood function parameters are used to fill in high-dimensional gradients and compensate for the compressed model's long-term information loss.
[50] --2021	Learned Gradient Compression for Distributed Deep Learning	Autoencoders are used to learn gradients for distributed hierarchical training and compression.
[51] --2022	Gradient Compression and Gradient Structure Correlation	Stochastic gradient correlations are retrieved and high-dimensional gradients are efficiently compressed through wireless aggregation.
[52] –2023	Federated Stochastic Gradient Descent with Mini-Batch	Jointly optimizes compression rate, batch size, and spectrum allocation.
[53] –2023	Heterogeneous Aware Gradient Compression	Client selection and gradient compression decisions are tightly coupled.
[54] –2024	Model Gradient Pruning	Low-complexity joint adaptive pruning and vehicle-user scheduling.
[55] –2024	Soft Clustering and 1-bit Compressed Sensing Federated Learning	An adaptive threshold mechanism based on 1-bit compression is used to reduce uplink transmission load.

2) Based on adaptive vehicle cluster research, we propose a dynamic gradient compression strategy, which sets an adaptive sparse threshold. The strategy reduces the participation of similar models in local aggregation and decreases communication costs.
3) We explore the mathematical problem involving dynamic gradient compression factors, converting the threshold issue into a statistical analysis challenge. Accordingly, we detail the similar model's calculation and optimization processes and derive its formula.
4) Finally, we verified our method's effectiveness using two general datasets, six traffic scenarios, and three comparative experiments with existing baseline methods [37], [47]. Our evaluation is based on four key aspects: a) model aggregation accuracy; b) convergence speed; c) compression ratio; and d) communication noise and interference resistance. The proposed gradient compression method achieves superior performance compared to the baselines.

This article is organized as follows. We describe the adaptive vehicle cluster construction and the FL system model in Section II, and introduce the gradient compression process. In Section III, we describe methods for calculating and optimizing dynamic gradient compression factors. Section IV provides the experimental results and analysis, and we conclude this article in Section II.

II. SYSTEM MODEL

This section presents a comprehensive overview of the FL framework and the gradient compression process during model aggregation. During gradient compression, we define the dynamic compression factor and describe the relevant theoretical symbols, as presented in Table II.

A. Adaptive Vehicle Cluster Construction

When a vehicle travels in a complex traffic area, it is selected as the roadside base station and establishes a connection through periodicity. After a connection has been established between the roadside base stations and vehicles, an adaptive vehicle cluster U belonging to the roadside base station BS_a is constructed according to three multiobjective

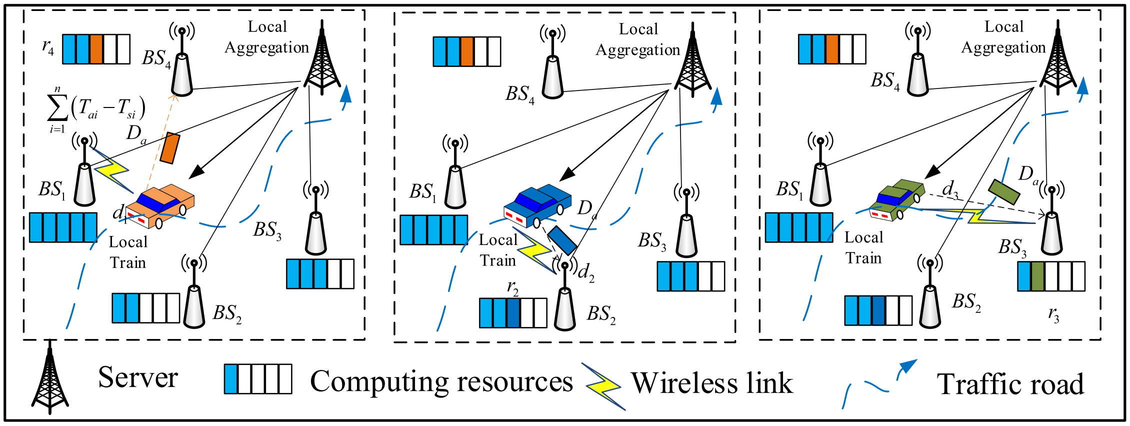

Fig. 2. Adaptive vehicle cluster construction mechanism.

TABLE II
SYMBOLS

Symbol	Define
d	Vehicle dataset sample nodes (horizontal volume)
l	Sample labels (column vector)
Y	Number of clients in the adaptive vehicle cluster
U	Adaptive vehicle cluster
D_i	Vehicle client sample collection
B_i	Min-batch
k	Local model training rounds
γ	Learning rate
λ_i	Model weight coefficient
s_{ij}	Model parameter probability distribution similarity
β	Base station preset communication distance weight

constraints: 1) the remaining resources; 2) the communication distance between the vehicle and the station; and 3) the latency from the vehicle to the roadside. Given the roadside base station set A; for BS_a, $a = 1, 2, \ldots, A$. Then, the vehicle identifies and calculates the base station's remaining communication resources r_a, including the distance d_a between the vehicle and the roadside base station and the average end-to-end delay D_a. Finally, the score Socre_a is calculated according to the three constraints of the vehicle and roadside base stations, and an adaptive vehicle cluster is constructed according to

$$\text{Socre}_a = \frac{r_a}{R} + \beta \times \frac{\text{Dis} - d_a}{\text{Dis}} + D_a \tag{1}$$

$$D_a = \frac{\sum_{i=1}^{n}(T_{ai} - T_{si})}{n} \tag{2}$$

where R is the maximum value of the remaining communication resource, Dis is the preset communication distance, and β is the base station's preset communication distance weight. In this instance, β controls the influence of the preset and actual communication distances on the score. When the distance between the vehicle and the base station is long, the $[(\text{Dis} - d_a)/\text{Dis}]$ will be large, resulting in poor communication quality and affecting the score. A larger β value indicates a greater communication distance impact on the score. A smaller β implies a reduced communication distance impact. Given that the vehicle nodes are from i to n, the time required for a sensing data group in the base station to leave the BS_a is T_{si}, and the time needed for arrival at the vehicle receiving node is T_{ai}. Then, the group's end-to-end delay is $D = T_{ai} - T_{si}$, and the average delay is D_a. Hence, the vehicle selects the roadside base station with the maximum Socre_a, establishing a connection.

Moreover, the optimal vehicle clusters are preferentially constructed using multiobjective constraints while conducting FL training and aggregation in intervals. As vehicle U_A moves, it gradually approaches base station BS_1, but the communication resources of BS_1 are fully utilized. U_A selects BS_4 to train a local model, which is suitable for dynamic traffic scenarios. Fig. 2. depicts the adaptive vehicle cluster's selection process for the base station's communication redundancy resources and traffic flow. In the illustration, the adaptive vehicle cluster U_A is close to BS_1, and the communication resources in that area are fully utilized. As a result, U_A switches its connection to BS_4 to upload its local model parameters and complete the aggregation process. U_1, U_2, and U_A vehicles on the road include: the distance of the vehicle identification roadside base station, the remaining computing resources, and the communication delay. 3-D attributes, which are multiobjective constraints for building adaptive vehicle clusters. According to Fig. 3(a) and (b), U_2 faces a communication resources shortage and is assigned the closest BS_2 to facilitate training by uploading its local parameters. Similarly, U_1 selects BS_3, providing abundant resources for local aggregation. The vehicle is observed in a highly dynamic driving situation with multiple vehicles in Fig. 3. The model is compressed to maintain robustness.

B. Dynamic Gradient Compression Factor

With the gradient compression factor δ_k as the sparsification edge, the dynamic factor is used as the nonfixed sparse threshold. However, this reduces the gradient parameter transmission for similar models and the interaction frequency between the vehicle and road terminals. It balances the computing resources during the aggregation process within the vehicle cluster. For each vehicle v_i in the adaptive cluster, the historical model gradient parameter distribution p_i^k is determined based

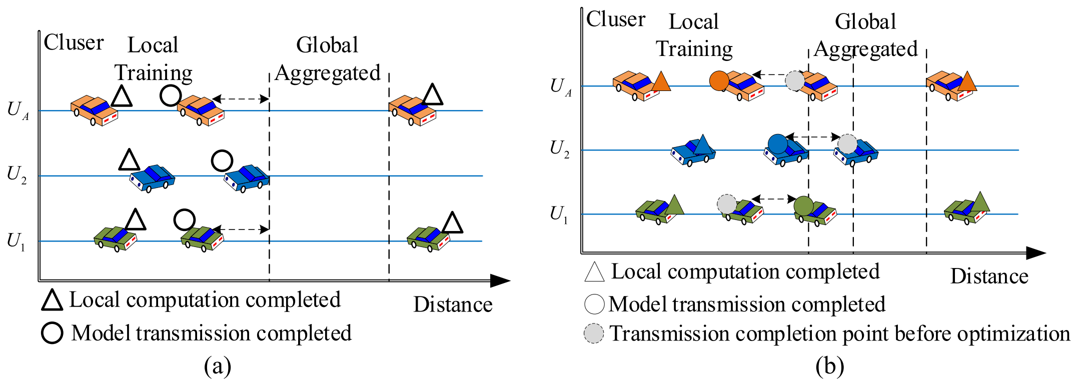

Fig. 3. FL using adaptive vehicle clusters. (a) Vehicle clusters U_1, U_2,..., U_A are trained locally, and the parameters are transferred for global aggregation. (b) Based on the dynamic gradient compression factor, the U_2 cluster is allocated more communication resources. Meanwhile, U_1 and U_A are allocated fewer resources to meet the asynchronous aggregation process's requirements.

on the local parameter set. This distribution is uploaded to the central parameter server at the roadside base station. Then, each vehicle client's weight coefficient in the aggregation model is calculated, and the results are used to select gradients for local aggregation. For the vehicle v_i in the cluster, the Wasserstein distance $\omega[p_i^{k-1}, q_j^{k-1}](i, j = 1, 2, \ldots, M;\ i \neq j)$ between the historical model gradient parameter distribution p_i^{k-1} and other distributions q_j^{k-1} is calculated. The Wasserstein distance measures the work required to transform one probability distribution into another. Specifically, it measures the minimum cost required to transform one distribution to another under the optimal transportation method [56], [57]. It is used to quantify the difference between each vehicle client's local and global model gradient distribution during FL. This distance facilitates the selection of gradients that make large contributions to local aggregation completion, optimizing the model training process and avoiding redundant data transmission. We assume that the number of vehicles is n, and we can calculate the distance between the historical model gradient parameter distribution and other distributions through the Wasserstein distance. It is used to calculate and select the minimum Wasserstein distance value, which is used as the gradient sparsification threshold. Finally, the minimum Wasserstein distance value is selected, and the current round's gradient compression factor δ_k is formulated as

$$\delta_k = \left[\min\left(\omega\left[p_i^{k-1}, q_j^{k-1}\right]\right)\right]. \tag{3}$$

Due to real-time traffic flow, varying computational capabilities, and heterogeneous communication resources within vehicle clusters, local training time differs across various lanes and vehicle types. Generally, clusters are constructed using dynamic vehicle nodes. As a result, model parameter transmission to roadside base stations may not be synchronized. Assuming that during the IoV's FL, there are several adaptive vehicle clusters $(U_1, U_2, \ldots, U_A)$, as shown in Fig. 3, the vehicle dynamics are reflected in two aspects.

1) U_2 is allocated more resources, and U_1 and U_A are assigned with less. This dynamic resource allocation meets the requirements for the asynchronous aggregation of vehicles in traffic scenarios.
2) The vehicle dynamics arise from asynchronous aggregation. Therefore, the adaptive vehicle cluster is constructed based on driving, leaving the cluster from U_2, and entering U_1.

Moreover, models in adaptive clusters are asynchronously aggregated, and vehicle actions and decision-making behaviors within various clusters vary depending on the base station's traffic flow and communication resource redundancy. Therefore, the dynamic compression factor (i.e., the gradient sparsification threshold) is optimized and calculated using the different adaptive vehicle clusters. Notably, optimizing the dynamic compression factors δ_k using various vehicle clusters is necessary. This is because the factor can control the gradient compression sparsity, thereby meeting the specific budget requirements. Furthermore, gradient compression can reduce the transmission parameters, addressing the limited and unbalanced nature of communication resources.

C. Local Gradient Compression and Upload

During training, the minimum set of data samples B_i is selected in the local dataset D_i, and the model is updated using the stochastic gradient descent (SGD) algorithm. This algorithm is used to perform weight vector updates for the vehicle's local model. Then, the updated weight vector $w_k^{j,\alpha}$ is obtained $(k = 0, 1, \ldots, K)$. In this instance, K is the FL local model training round, and the gradient parameters after each weight vector update are calculated as $F(w_k^{i,\alpha}; B_i)$. Hence, the update process for the weight vector $w_k^{j,\alpha}$ is formulated as

$$w_k^{i,\alpha} = w_k^{i,\alpha-1} - \gamma F\left(w_k^{i,\alpha-1}; B_i\right) \tag{4}$$

where $F(w_k^{i,\alpha}; B_i)$ is the global gradient parameter $\bar{F}(w_{k-1}, D)$ obtained by the $k-1$th FL local training round, $w_k^{i,\alpha}$ is the client of vehicle v_i, which is a weight vector updated according to the global gradient parameters, and γ is the preset learning rate. In this scenario, the model gradient parameters $F(w_k^{i,\alpha}; B_i)$ calculated in this training round are uploaded to the roadside base station involved in local gradient compression. Hence, the local aggregation compression gradient's kth vehicle client training round is expressed as

$$F\left(w_k^i\right) = \sum_{\alpha=1}^{\alpha} \delta_k F\left(w_k^{i,\alpha}; B_i\right). \tag{5}$$

In this context, δ_k is used as the FL process's local aggregation sparse gradient compression factor. Specifically, gradients with significant contributions are selected and locally aggregated, reducing the repetition of transmissions for similar models. As a result, the communication cost of local model aggregation within the vehicle cluster is reduced. For each v_i in the adaptive vehicle cluster, the historical model gradient parameter distribution p_i^k is determined based on the local set. This distribution is uploaded to the central parameter server at the roadside base station. Then, the weight coefficient of each vehicle client in the model is calculated and used to select gradients for local aggregation. For the cluster's v_i, the calculated Wasserstein distance between vehicles is used as the dynamic compression factor δ_k distributed by the historical model's gradient parameters, which is used as the distribution similarity s_{ij}. Thus, the weight coefficient φ_{ij} is calculated using the sigmoid function in

$$\varphi_{ij} = \frac{1}{1 + e^{-s_{ij}}}. \tag{6}$$

For this formula, the φ_{ij} range is within [0,1]. A larger distribution similarity measure s_{ij} implies that φ_{ij} is closer to 1, indicating a higher similarity level in the locally trained models. Typically, higher similarity models contribute more significantly to the global model during local aggregation, while lower ones have a relatively smaller impact on the updates. Therefore, when the sparsification method compresses gradients, the dynamically selected gradient compression factor δ_k is used as a sparse threshold to eliminate similar models. Furthermore, models with significant differences are retained, ensuring stable and robust global aggregation accuracy. Next, the model weight is quantified according to similarity, increasing the participation of gradient parameters with large contributions in the aggregation process. Finally, each participating vehicle's global model aggregation weight λ_i within the adaptive cluster is calculated based on the coefficient φ_{ij}, as described in

$$\lambda_i = \frac{1}{M-1} \sum_{j=1,2,\ldots,M, j\neq i} \varphi_{ij}. \tag{7}$$

For Algorithm 1's dynamic gradient compression, we describe the computational complexity as a subset sampled from the dataset D_i at each iteration. This complexity depends on the number of elements in the dataset with the sample number being proportional to the mini-batch size B_i. Furthermore, compression complexity is linearly related to gradient size. In addition, the compression operation size is M, and the complexity calculation is $O(M)$.

D. Global Model Update

The central parameter server calculates the compressed global gradient vector $\bar{F}(w_k, D)$ and model update using the following formula:

Algorithm 1 Dynamic FL Gradient Compression

Input:Dataset D_i, mini batch size B_i, the learning rate γ.
Output: Dynamic compression factor δ_k, global model $\bar{F}(w_k, D)$.

```
D_i ← 0 # Initialization process
for k = 0, 1, ..., max_iters do # Start the training process iteration
    for Sample (d_z, l_z) elements as w_k^{i,α−1};
        SGD(w, decompress(w_k^{i,α−1})) # SGD updating;
        w_k^{i,α} = w_k^{i,α−1} − γF(w_k^{i,α−1}; B_i);
        F(w_k^i) ←[Σ_{α=1}^{α} δ_k F(w_k^{i,α−1} B_i)]— all(compress(w_k^{i,α}))# Gradient compression;
    end for
        for φ_ij ← ω[p_i^k, q_j^k](φ_ij ∈ [0, 1]) do # Node Updating
        end for
    λ_i ← 1/M − 1 Σ_{j=1,2,...,M, j≠i} φ_ij # Calculate weighting factor;
    F̄(w_k, D) ← 1/M Σ_{i=1}^{M} λ_i F̄(w_k^i) # Global model aggregation;
end for
```

$$\bar{F}(w_k, D) = \frac{1}{M} \sum_{i=1}^{M} \lambda_i \bar{F}\left(w_k^i\right) \tag{8}$$

where D is the union of M vehicle local data sample sets D_i in the cluster U. The roadside base station's parameter server broadcasts the global gradient vector $\bar{F}(w_k, D)$ to the vehicles in cluster U for decision making and data sharing. Hence, the FL dynamic gradient compression calculation process is described in Algorithms 1 and 2.

In this instance, the initialization operation is a constant time $O(1)$, and the dataset size for each sampling round is B_i. The computational complexity is expressed as $O(B_i)$, where B_i is the mini-batch size processed by each vehicle's local model. Additionally, the computational complexity of the SGD update and decompression is $O(B_i + M)$, where M is the model size. The update and calculation of the mask selection and gradient compression's vehicle nodes within a cluster are φ_{ij} and λ_i, respectively, and the complexity of their calculation weighting factors is $O(M)$. For each iteration and compression of the global model update, the total complexity is expressed as

$$O(B_i + M). \tag{9}$$

In addition, the number of aggregation iterations of each FL is max_iters, and the algorithm complexity is formulated as

$$O(\text{max\_iters} \times (B_i + M)). \tag{10}$$

Algorithm 2 describes the dynamic gradient compression strategy for the FL of adaptive vehicle clusters in IoV. Its three key properties are as follows.

1) *Compressed Local Aggregated Gradients:* The strategy employs gradient compression techniques to reduce the size of the local aggregated gradients transmitted during FL.

Algorithm 2 FL-DGC of the Adaptive Vehicle Cluster

Input:Dataset D_i, mini batch size B_i, the learning rate γ,
Output: The BS broadcast w_{k+1}^{i},

1: Initialize the model parameters: w_0
2: Initialize dynamic gradient compression factor δ_k
3: for each cluster i in (1, 2, ..., M) do # Node updating;
4: Sample mini-batch B_i from D_i # Sample mini-batch;
5: $w_k^{i,\alpha} = w_k^{i,\alpha-1} - \gamma F\left(w_k^{i,\alpha-1}; B_i\right)$ # Perform SGD updating;
6: for $k \in \{1, 2, ..., K\}$
7: $F\left(w_k^i\right) = \sum_{\alpha=1}^{\alpha} \delta_k F\left(w_k^{i,\alpha}; B_i\right)$# Gradient compression;
8: The BS computes $\bar{F}\left(w_k^i\right)$;
9: The BS transmit $\bar{F}\left(w_k^i\right)$ to the server;
10: The BS averages $F(w_k, D) = \frac{\sum_{i=1}^{M} \lambda_i \bar{F}\left(w_k^i\right)}{|M|}$ from vehicles;
11: The BS updates the global model, the BS broadcast w_{k+1}^{i} to vehicles $U_1 \ldots U_M$;
12: end for
13: end for

2) *Dynamic Resource Allocations:* This strategy dynamically adjusts the allocation of communication resources within the adaptive vehicle cluster.
3) *Single Global Gradient Aggregation:* This simplifies the aggregation process and reduces the complexity of model updates while maintaining the aggregated model's accuracy and effectiveness.

III. Problem Formulation and Optimization Analysis

In this section, we elucidate the gradient compression factor's optimization process in the vehicle clusters. We formalized the gradient compression factor's calculation using mathematical formulas.

A. Calculation of Dynamic Compression Factor

During the model's distributed training, the vehicle clients employ compression to select a subset of gradients during each aggregation iteration. Within the deep neural network, the sparse threshold for the cumulative gradient w_k^i of each layer in the local model is computed based on the dynamic compression factor δ_k. Using δ_k, the cumulative gradient is compressed into a sparse global one $\overline{F}(w_k, \theta)$ for communication. Then, the sparsity threshold is calculated according to the dynamic compression factor, enabling selective matrix sparsification.

After the client selects and sparses the model's gradient, it is further compressed and uploaded to the central parameter server in the roadside base station. Within this station, the parameter server aggregates to obtain the global gradient $F(w_k, \theta)$, where θ is a global parameter proven to be stable with the time iteration, converging to a certain optimal solution θ^*. The gradient compression factor optimization is expressed as

$$\min_{\delta_k} E_{\{B_i\}}\left[\min_{k\in\{0,\ldots,K\}} F(w_k, \theta)\right]. \tag{11}$$

Expressed as the clients v_i and v_j in the vehicle cluster U, the model gradient parameters obtained by local training satisfy the distributions p_i and q_j, which have the same distribution properties. They are represented as the training model's gradient distribution for each vehicle. Then, the probability distribution between vehicles is updated dynamically in real time based on historical perception data. Next, the dynamic gradient compression algorithm is deployed in the IoV, and the source similarity of vehicle perception data within a cluster is considered and modeled.

Moreover, the close-range vehicles' perception data in the cluster are different samples within the same overall distribution, indicating that the vehicle data vary at different viewing distances. Therefore, $\omega[p_i, q_j]$ is defined to represent the Wasserstein distance between different clients, indicating that the model parameters' gradient distributions are similar. We calculate and optimize δ_k according to the statistical distribution of the gradient parameters for models trained by different clients in the adaptive vehicle cluster. The distance function (i.e., the transfer cost) is set to $d(x, y)$. In this instance, x indicates that the model gradient parameter of v_i satisfies $x \sim p_i$, and y shows that v_j satisfies $y \sim q_j$. Thus, the Wasserstein distance calculation is defined as

$$\omega\left[p_i, q_j\right] = \inf_{\mu_k\in T[p_i,q_j]} \iint \mu_k(x, y) d(x, y) d_x d_y \tag{12}$$

where $\mu_k \in T[p_i, q_j]$ is the joint distribution set of the gradient parameter distributions p_i and q_j. Hence, we use the following formula to calculate the Wasserstein distance of the two distributions:

$$\inf_{\mu_k\in T[p_i,p_j]} \iint \mu_k(x, y) d(x, y) dxdy \tag{13}$$

$$\begin{cases} \int \mu_k(x, y)dy = p(x) \\ \int \mu_k(x, y)dx = q(y) \\ \quad \mu_k(x, y) \geq 0. \end{cases} \tag{14}$$

In the formula, $p(x)$ represents the marginal distribution of the probability distribution $\mu_k(x, y)$ on the x-axis, with which the probability distribution of x when y is determined. Similarly, $q(y)$ is the marginal distribution of the probability distribution $\mu_k(x, y)$ on the y-axis, with which the probability distribution of y when x is determined. Furthermore, the dynamic network's topology complexity and continuous distribution in the IoV is an issue with countless decision variables and innumerable constraints. Assuming that the dual variable is a function $f(x)$, the communication resource constraint is a prerequisite for addressing the dual resources challenge. Therefore, transforming the multiplication of the dual variables and equality constraints into an integral calculation. Hence, we set the dual variable for the second constraint to be $g(x)$, and the function is computed as

$$\begin{aligned} &\inf_{\mu_k(x,y)\geq 0} \iint \mu_k(x, y) d(x, y) dxdy \\ &\quad + \int f(x)\left[p(x) - \int \mu_k(x, y)\mathrm{dy}\right]\mathrm{dx} \\ &\quad + \int g(y)\left[q(y) + \int \mu_k(x, y)\mathrm{dx}\right]\mathrm{dy}. \end{aligned} \tag{15}$$

Then, we obtain the formula

$$\inf_{\mu_k(x,y)\geq 0} \int f(x)p(x)dx + \int g(y)q(y)\mathrm{d}y + \iint \mu_k(x,y)\big[d(x,y) - f(x) - g(y)\big]dxdy. \tag{16}$$

If $d(x, y) - f(x) - g(y) \leq 0$, proper $\mu_k(x, y) \geq 0$ is obtained, bringing (11) close to an infinitesimal derivation of the dual optimization problem, as displayed in

$$\sup_{f,g} \int \big[f(x)p(x) + g(x)q(x)\big]dx \tag{17}$$

$$\omega\big[p_i, q_j\big] = \sup_{f,g}\left\{\int \big[f(x)p(x) + g(x)q(x)\big]\mathrm{d}x\right\} \quad \text{s.t.}|f(x) + g(y) \leq d(x, y). \tag{18}$$

In addition, the transition cost is defined as $d(x, y) = 0$. Therefore, (19) can be calculated by setting $g(x) = -f(x)$

$$\omega\big[p_i, q_j\big] = \sup_{f}\left\{\int \big[p(x)f(x) + q(x)g(x)\big]\mathrm{d}x\right\} \quad \text{s.t.}|f(x) - g(y) \leq d(x, y). \tag{19}$$

Generally, $d(x, y) = \|x - y\|_p$ is selected as the Euclidean space's p-norm, indicating that it is the optimal solution. Furthermore, the δ_k-optimal solution for (12) and (20) is transformed into the joint optimization

$$\min\big(p_i, q_j\big) = \sup_{\mu_k\sim T(p_i,q_j)} E_{(x,y)\sim\mu}\big[\|x - y\|\big] \rightarrow \min_{\{\sigma_k\}} E_{\{B_j\}} \min_{\delta_k} E_{\{B_j\}}\left[\min_{k\in\{0,\dots,K\}} F(w_k, \theta)\right]. \tag{20}$$

B. Convergence Analysis

During FL, the SGD update is used to rule for convergence proof, where r_k is the learning rate at the kth step, as described in

$$\theta^{(k+1)} = \theta^{(k)} - r_k\bar{F}_k \tag{21}$$

where f is the gradient compression result after Wasserstein distance control. According to the approximation of the expected gradient, we obtain

$$\mathbb{E}\big[\bar{F}_k\big] = \nabla f\Big(\theta^{(k)}\Big). \tag{22}$$

In this scenario, the sparse threshold is determined based on the Wasserstein distance, and the gradient is compressed. Assuming the errors ε_k and $\varepsilon_k \rightarrow 0$ change with the number of iterations k, we can formulate

$$\mathbb{E}\big[\bar{F}_k\big] = \nabla f\Big(\theta^{(k)}\Big) + \varepsilon_k. \tag{23}$$

Then, the SGD is used to update the model according to

$$\theta^{(k+1)} = \theta^{(k)} - r_k\Big(\nabla f(\theta^{(k)}) + \varepsilon_k\Big). \tag{24}$$

Next, it is expanded as $\mathbb{E}[\|\theta^{(k+1)} - \theta^*\|^2]$

$$\mathbb{E}\Big[\|\theta^{(k+1)} - \theta^*\|^2\Big] = \mathbb{E}\Big[\|\theta^{(k)} - r_k\Big(\nabla f(\theta^{(k)}) + \varepsilon_k\Big) - \theta^*\|^2\Big]$$

$$\mathbb{E}[\|\theta^{(k+1)} - \theta^*\|^2 = \mathbb{E}\Big[\|\theta^{(k)} - \theta^*\|^2\Big] \tag{25}$$

$$-2r_k\mathbb{E}\Big[\langle\theta^{(k)} - \theta^*, \nabla f\Big(\theta^{(k)}\Big) + \varepsilon_k\rangle\Big] + r_k^2\mathbb{E}\Big[||\nabla f\Big(\theta^{(k)}\Big) + \varepsilon_k||^2\Big]. \tag{26}$$

To continue analyzing the update rule's effect on the convergence of parameter θ, the term $\nabla f(\theta^{(k)})$ is processed using the expectation of the stochastic gradient $\mathbb{E}[\langle\theta^{(k)} - \theta^*, \nabla f(\theta^{(k)})\rangle]$. According to the function convexity definition for all θ, θ^*, and random gradient $\nabla f(\theta^{(k)})$, the following formulas can be obtained:

$$f\big(\theta^*\big) \geq f\Big(\theta^{(k)}\Big) + \langle\nabla f\Big(\theta^{(k)}\Big), \theta^* - \theta^{(k)}\rangle \tag{27}$$

$$\langle\nabla f\Big(\theta^{(k)}\Big), \theta^{(k)} - \theta^*\rangle \geq f\Big(\theta^{(k)}\Big) - f\big(\theta^*\big). \tag{28}$$

Therefore, the expected value is satisfied

$$-\mathbb{E}\Big[\langle\theta^{(k)} - \theta^*, \nabla f\Big(\theta^{(k)}\Big)\rangle\Big] \leq -\Big(f(\theta^{(k)}) - f(\theta^*)\Big). \tag{29}$$

Assuming that the term ε_k is introduced through gradient compression and the error expectation is small (i.e., the error decreases as the number of iterations increases). In the scenario of gradient compression, the error expectation is small, which means that although errors ε_k are introduced by compression during the training process, these errors do not lead to a serious degradation of model performance, and the model training results still converge, we can obtain the formula

$$\mathbb{E}\Big[\langle\theta^{(k)} - \theta^*, \varepsilon_k\rangle\Big] \approx 0. \tag{30}$$

In this equation, we assume that the second moment of ε_k is bound (i.e., $\mathbb{E}[\|\varepsilon_k\|^2] \leq C$). Furthermore, the gradient norm expectation can be denoted as

$$\mathbb{E}\Big[\|\nabla f\Big(\theta^{(k)}\Big) + \varepsilon_k\|^2\Big] \leq 2\mathbb{E}\Big[\|\nabla f\Big(\theta^{(k)}\Big)\|^2\Big] + 2\mathbb{E}\Big[\|\varepsilon_k\|^2\Big]. \tag{31}$$

Due to $\|\nabla f(\theta^{(k)})\| \leq G\,\|\varepsilon_k\|^2 \leq C$, we can formulate

$$\mathbb{E}\Big[\|\nabla f\Big(\theta^{(k)}\Big) + \varepsilon_k\|^2\Big] \leq 2G^2 + 2C. \tag{32}$$

Therefore, by integrating the above results into the convergence analysis, we can obtain

$$\mathbb{E}[\|\theta^{(k+1)} - \theta^*\|^2 = \mathbb{E}\Big[\|\theta^{(k)} - \theta^*\|^2\Big] -2r_k\Big(f(\theta^{(k)}) - f(\theta^*)\Big) + r_k^2\Big(2G^2 + 2C\Big). \tag{33}$$

In this formula, when the learning rate is r_k, $\mathbb{E}[\|\nabla f(\theta^{(k+1)}) + \varepsilon_k\|^2]$ approaches 0, indicating that the model parameter θ converges to the global optimal θ^*.

IV. Experimental Results

The experiments were conducted on a hardware system equipped with an Intel Core i7-11700F CPU and NVIDIA RTX3060Ti GPU. The simulation was implemented using the PyTorch framework (version 1.1.0, CUDA 11.3) to build a distributed training platform. The GPU and CPU communication was facilitated using PCI-E 4.0.

This article's simulation aims to investigate the impact of dynamic compression factors on FL performance in different

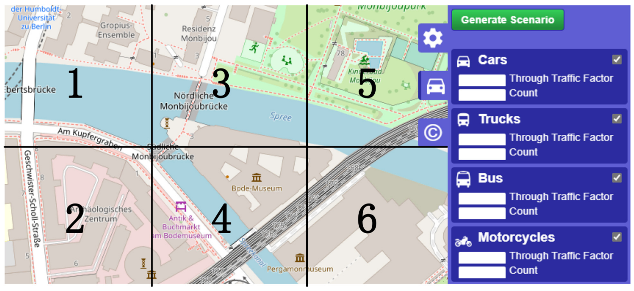

Fig. 4. Clusters and aggregation groups according to the adaptive vehicle cluster.

TABLE III
ADAPTIVE VEHICLE CLUSTER PARAMETERS AND SETTINGS

Clusters Number	Aggregation Groups	Number Vehicles	Communication Delay	Average Speed
1	7	50	15 Mbit/s	40 km/h
2	10	40	10 Mbit/s	30 km/h
3	2	30	20 Mbit/s	50 km/h
4	6	20	40 Mbit/s	100 km/h
5	2	10	60 Mbit/s	60 km/h
6	4	20	50 Mbit/s	100 km/h

spatial scopes and traffic environments. We used SUMO to generate traffic maps, which were further divided into six clusters. The maps were generated by osmWebWizard in the SUMO 1.8.0 version, and we used different clusters in WebWizard to evaluate the vehicle aggregation model's performance. SUMO generates maps that satisfy and adapt to our real traffic scenarios as much as possible, which is a common approach in traffic. The generated traffic map encompasses an area in Berlin, Germany, including parks, residential and commercial areas, highways, rivers, and bridges. Diverse traffic scenario simulations are used to verify the scalability of the strategy. In the IoV, clusters must share and merge perception data collected from multiple sensors installed on roadside bases and vehicle clients. These data serve as input for FL tasks such as local path planning for congested road sections.

We use these two types of dataset simulations, which can evaluate the performance of the model from both grayscale and RGB perspectives. In addition, they are used to simulate different types of image data collected by vehicles to perform relevant model training. Our strategy employs the CIFAR-10 and MNIST datasets to train and evaluate the test image classification performance. Specifically, the MNIST dataset contains 70 000 images from ten classes, including 60 000 training and 10 000 testing samples. The data type of the MNIST dataset is a grayscale image. It is an image dataset of handwritten digits and it is often used for benchmarking image classification and learning models. The CIFAR10 dataset comprises 42 000 vehicle-related RGB images, such as cars and trucks, selected for validation. The datasets do not contain any overlapping data, meeting the independent and identical distribution requirements. The CIFAR-10 dataset is mainly of RGB image type. It is a standard dataset for image classification and contains color images of ten different categories. That is widely used for image classification. This allows us to evaluate the adaptive vehicle clustering's gradient compression performance in each region using various dynamic methods [37], [47].

Based on the adaptive vehicle clusters' 3-D attribute construction mechanism, as shown in Fig. 4, we perform three sets of experiments using various parameters, which compare the model's performance in each cluster. According to Fig. 4. and Table III, the number of vehicle aggregation groups and the average number of vehicles in each cluster are larger in clusters 1 and 2 (residential). In this traffic scenario, vehicles move at a slow speed with a high data transmission latency. Clusters 1 and 2 are configured with 50 and 40 vehicles. Their model aggregation times are the largest, which are 7 and 10 times. In contrast, cluster 5 (Parks), which is the greenest, has a low gathering frequency and a small number of vehicles in each group. The vehicles experience low communication latency relative to their high speeds. Comparing clusters 3 (Parks and Commercial) and 5 indicates that a difference in the number of vehicles exists despite having the same number of aggregations. Cluster 5's latency increases, as the average vehicle speed decreases. There are 30 and 10 vehicles in vehicle clusters 3 and 5, and their aggregation model times are the same, which is set to 2 times. Similarly, in clusters 4 (Highway and Bridges) and 6 (Highway and Commercial), a significant difference in aggregation frequencies exists although the number of vehicles is the same; however, a small difference in latency occurs. In vehicle clusters 4 and 6, the number of vehicles simulated in the scenarios is 20 vehicles, and the number of model aggregations is 4 and 6 times.

To evaluate the dynamic gradient compression strategy's implementation performance in adaptive vehicle cluster variability, we designed baseline methods, including the Top-K [37] and LGC [47] for comparative experiments according to the application requirements. The Top-K selects the largest k gradients from the sorted ones, which is a type of sparsification. In contrast, LGC uses local compression quantization to compress the gradient, reducing the number of bits required to represent it.

Experiment Group 1: This involves the compression model stability assessment. Other traffic participants introduce noise in complex traffic scenarios, which disturbs the steady-state nature of the model aggregation process. We use signal-to-noise ratio (SNR), which is used to evaluate the difference between the original model and the compressed model. SNR can help quantify the degree of information loss during the compression process. The higher the SNR, the better the signal quality, and the smaller the difference between the compressed model and the original model, and vice versa. In addition, the strategy's stability and noise resistance capacity in the group 1 experiment are prerequisites for experimental groups 2 and 3. Therefore, adequate stability and reliability allow meaningful model aggregation.

Experiment Group 2: This is a qualitative analysis of the convergence evaluation. These experiments aim to evaluate the gradient compression strategy model's convergence training and aggregate model accuracy in various adaptive vehicle clusters. Federated aggregate model accuracy, which is used

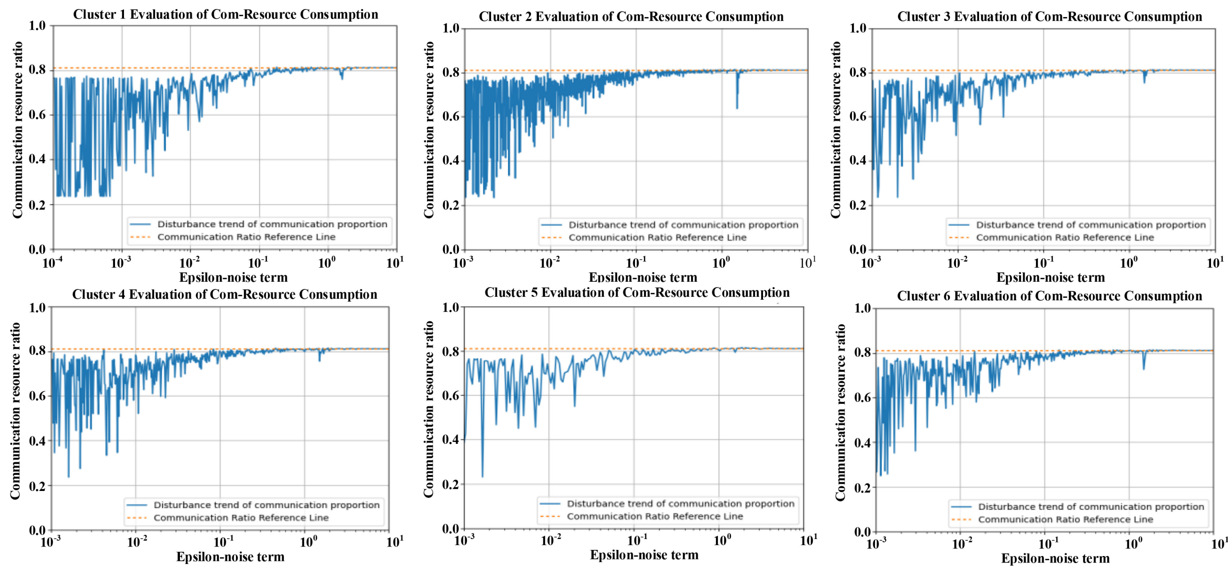

Fig. 5. Simulation of communication resource proportions and disturbances within different clusters.

to evaluate the final performance of the model on the task. We set the loss function curve for qualitative analysis. If the loss value keeps decreasing, it means that the model is continuously optimizing; if the decrease tends to be gentle, it means that the model is close to the optimal solution. The results directly demonstrate that the proposed method can achieve convergence based on the FL architecture within the IoV network's topology.

Experiment Group 3: This is a quantitative analysis of the comparative performance evaluation. We set the compression ratio, which is used to validate and evaluate the performance of the method compared with the baseline method in the interaction of the number of model parameters. These experiments prove that the proposed strategy surpasses the other two gradient compression methods in convergence speed and aggregation accuracy.

A. Group 1: Compression Model Stability Assessment

The strategy's stability and noise resistance capability in the group 1 experiment are prerequisites for groups 2 and 3. Therefore, adequate stability and reliability allow meaningful model aggregation. For the practical application requirements, we considered the vehicle clusters' limited communication resources. Meanwhile, noise was introduced by other traffic participants in heterogeneous traffic scenarios, a stable process that can disturb the communication network. During the simulation, a random noise term (epsilon) was added to generate simulation labels [37], [47]. The epsilon term follows a normal distribution with a mean of 0 and a standard deviation of 0.01. Noise is meaningless interference in the dataset, and the communication resource ratio reference line is set to 80% ($\pm$1%). The simulation was conducted to demonstrate FL-DGC's steady state and convergence process when noise interference occurs across each cluster, as shown in Fig. 5.

In cluster 1, which is a mixed residential area, the vehicles and vehicle aggregation groups occupy a significant proportion of the communication resources. Therefore, more disturbances occur, resulting in a long stabilization process for the proposed strategy. Cluster 2 includes a mix of residential areas and main roads. This includes the number of vehicles and vehicle aggregation groups, which account for the largest proportion of communication resources. Therefore, the noise disturbance is the most prominent, resulting in a longer stabilization process than cluster 1.

In clusters 3 and 5, the aggregation groups in the model remain the same while the vehicle number varies. Fig. 5 demonstrates that a small number of vehicles results in less noise interference. For clusters 4 and 6, the vehicle numbers are the same, but the number of times the models are aggregated differs. Fig. 5 indicates that a small number of aggregation groups in 4 results in a higher convergence rate than the aggregated model in cluster 6.

Consequently, the proportion of communication resources allocated to cluster 4 is smaller than 6. Accordingly, we conclude that dynamic compression factors can vary according to the number of vehicles in the environment, and are influenced by the participating vehicle density within the vehicle clusters. Despite varying noise influences, the proposed dynamic compression strategy demonstrates stable convergence in adaptive vehicle clusters. In each cluster, the strategy can effectively adjust the allocation of communication resources within the vehicle clusters, ensuring efficient utilization and minimizing waste. The proposed dynamic gradient compression strategy's stability and reliability were verified, laying the foundation for the experiments in groups 2 and 3.

During the gradient compression integrity and anti-interference experiments, the SNR indicator was utilized to assess variations in the model before and after compression under various conditions and using different baseline methods. Specifically, SNR measures the signal strength relative to that of the background noise. For gradient compression, the signal refers to the gradient's useful information and noise is the error or distortion introduced by compression. SNR is measured in

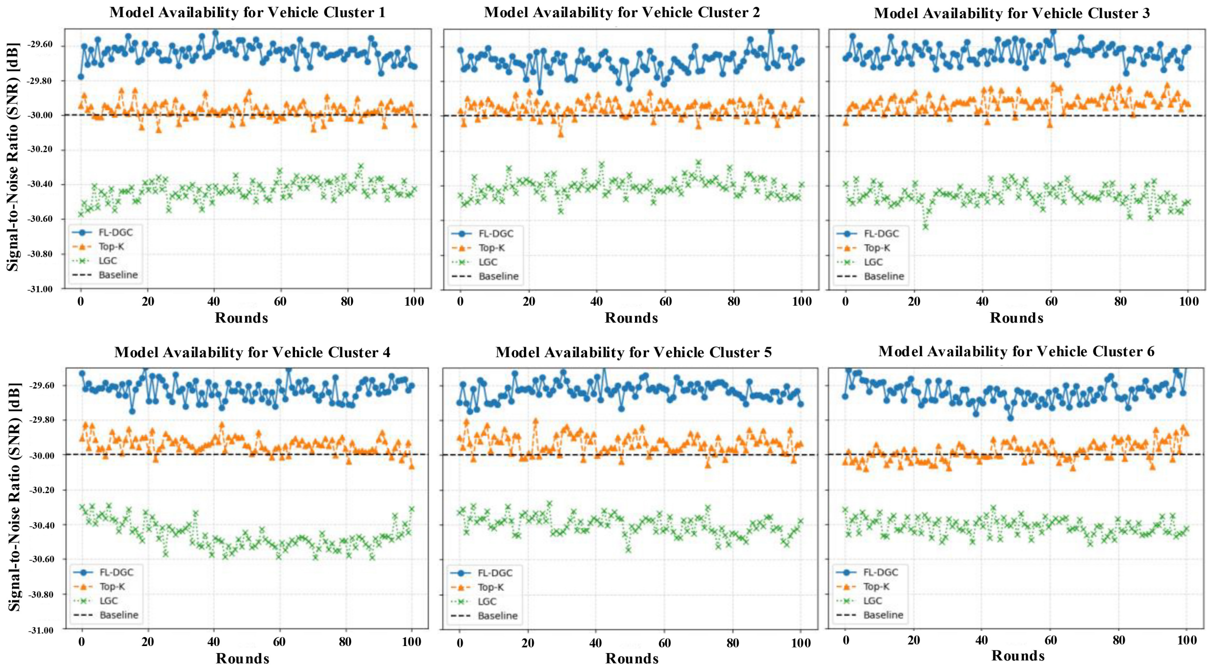

Fig. 6. Comparison of gradient compression SNR for different vehicle cluster models.

decibels (dB), providing a logarithmic quantitative indicator for signal quality, as shown in

$$SNR = 10\log_{10}\left(\frac{\|\nabla\|^2}{\|\nabla - \widehat{\nabla}\|^2}\right). \tag{34}$$

In (34), ∇ is the original model's parameter and $\widehat{\nabla}$ is the compressed one. Meanwhile, $\|\nabla\|^2$ is the energy of the original signal and $\|\nabla - \widehat{\nabla}\|^2$ represents that of the error. When the error is small, ∇ and $\widehat{\nabla}$ are close and the SNR value is high, indicating improved signal preservation and less noise. According to Fig. 6, we observe that FL-DGC maintained the availability and integrity of the original model's parameters. In addition, FL-DGC always maintains the highest SNR, implying that the information variation between the compressed and original models is small in different vehicle clusters. The SNR of FL-DGC is maintained at -29 dB$\pm$0.1, and the LGC local gradient compression has a lower SNR. Vehicle cluster 2 has a larger DGC fluctuation due to the increase in vehicles and aggregation times. In clusters 3 and 5, we observe that the LGC algorithm is more volatile in cluster 5, while FL-DGC is more stable in cluster 5. In summary, FL-DGC is effective in ensuring model availability.

B. Group 2: Convergence Evaluation

The experiments in group 2 demonstrated that the FL-DGC strategy generates a superior global model after aggregation. We utilized the CIFAR-10 dataset and allocated varying amounts of data (i.e., n/42 000, where n= 20 000, 30 000, and 40 000) based on actual cluster divisions. Our baseline for achieving a model accuracy of 80% ($\pm$1%) was set using sparsity. Group 1 specifically aimed to validate the global model's convergence rate using the FL-DGC strategy in different clusters. During each iteration cycle (i.e., epoch), we ensure a complete traversal of the *data_iter* function, which is utilized to process all the samples in the training dataset (assuming the number of samples is divisible by the batch size). According to the actual implementation requirements, we set the epochs for each cluster to 200. The experiment included six clusters, each with a different number of aggregation groups and allocated vehicles per group. Then, we monitored the average model accuracy and loss function value curves for clusters 1–6 using three methods. Hence, this experiment primarily uses the sample increment method.

1) Clusters 1 and 2 have a larger number of traffic participants and aggregation groups than 3 to 6, and the perception data was used for the classification task. The analysis demonstrated that the FL-DGC method achieves faster convergence and improved stability in complex urban traffic environments, such as residential areas and major traffic arteries. As shown in Fig. 7, the aggregation model trained using FL-DGC exhibited seamless convergence and stability at approximately 100–125 epochs, compared to other scenarios. The aggregated model's median accuracy is 92.6% and 93.2%, indicating the FL-DGC strategy's efficiency. This also demonstrates that FL-DGC leads to higher model timeliness than the other two gradient compression methods.

Meanwhile, we analyzed the results in cluster 1, as shown in Table IV. The dynamic gradient compression factor in cluster 1 was adjusted when the data increased by 40 000/42 000, improving the model's global aggregation accuracy. These qualitative analyses demonstrated the comparison between our strategy and the other two fixed gradient sparsification methods in different clusters. They show that the dynamic gradient factor enhances the FL-DGC strategy's robustness and generalizability.

2) Clusters 3 and 5 have equal aggregation groups, but the vehicle number within each cluster differs.

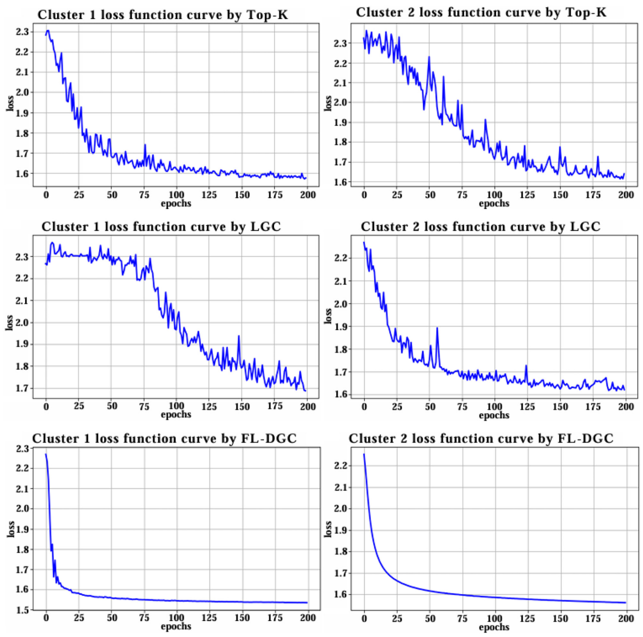

Fig. 7. Simulation comparison of clusters 1 and 2.

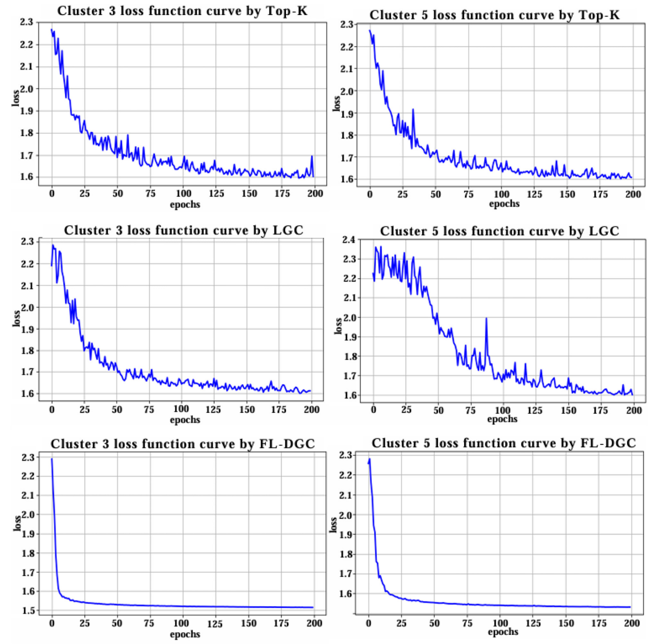

Fig. 8. Simulation comparison of clusters 3 and 5.

TABLE IV
DISTRIBUTED TRAINING ACCURACY AND LOSS USING DIFFERENT INFORMATION

Project	Average Acc	Loss	Inf
Baseline	80% (±1%)		Inf size change
Cluster 1 -Top-K	79.4% −0.6%	2.012	40,000/42,000
Cluster 1 -LGC	66.3% −13.7%	3.681	40,000/42,000
Cluster 1 -DGC	92.6% +10.6%	1.338	40,000/42,000
Cluster 2 -Top-K	80.4% +0.4%	1.612	40,000/42,000
Cluster 2 -LGC	86.3% +6.3%	1.570	40,000/42,000
Cluster 2 -DGC	93.2% +13.2%	1.212	40,000/42,000

TABLE V
DISTRIBUTED TRAINING ACCURACY AND LOSS USING DIFFERENT INFORMATION IN CLUSTER 1 AND 2

Project	Average Acc	Loss	Inf
Baseline	80% (±1%)		Inf size change
Cluster 3 -Top-K	86.1% +6.1%	1.756	20,000/42,000
Cluster 3 -LGC	84.3% +4.30%	2.133	20,000/42,000
Cluster 3 -DGC	93.8% +13.8%	1.355	20,000/42,000
Cluster 5 -Top-K	85.7% +5.7%	2.261	20,000/42,000
Cluster 5 -LGC	79.1% −0.90%	3.371	20,000/42,000
Cluster 5 -DGC	87.6% +7.60%	1.536	20,000/42,000

In the image classification vehicle cluster, the accuracy and loss function of 3 and 5 were compared using different information, as shown in Table V. Clusters 3 and 5 represent mixed traffic environments consisting of buildings and parks, and the number of aggregation groups and vehicles are different. According to Fig. 8, the Top-K and LGC values tend to converge and plateau within 84–100 epochs, while the FL-DGC strategy converges within 50 epochs. We compared the aggregation model's convergence speed in clusters 3 and 5, and the results were consistent with the actual traffic scenario. According to the analysis, the aggregated model's average accuracy in cluster 3 is higher than 5. Specifically, the aggregated model's accuracy in cluster 3 is 93.8%, which is 6.2% higher than 5.

TABLE VI
DISTRIBUTED TRAINING ACCURACY AND LOSS USING DIFFERENT INFORMATION IN CLUSTER 3 AND 5

Project	Average Acc	Loss	Inf
Baseline	80%(±1%)		Inf size change
Cluster 4 -Top-K	84.1% +4.1%	2.306	30,000/42,000
Cluster 4 -LGC	83.7% +3.7%	2.298	30,000/42,000
Cluster 4 -DGC	94.4% 14.4%	1.263	30,000/42,000
Cluster 6 -Top-K	86.7% +6.7%	1.913	30,000/42,000
Cluster 6 -LGC	73.3% -6.7%	3.816	30,000/42,000
Cluster 6 -DGC	95.0% +15.0%	1.221	30,000/42,000

3) Clusters 4 and 6 have the same number of vehicles but different aggregation groups. We compared and analyzed the results displayed in Table VI.

Clusters 4 and 6 represent pure traffic environments with buildings and main roads. Compared to the previously mentioned groups, these two clusters exhibited differences in real-time vehicle flow and communication resource redundancy. In cluster 4, LGC tends to converge and plateau within 125 epochs, and the Top-K within 175. Meanwhile, FL-DGC converged at approximately 75 epochs. Similarly, when comparing and analyzing the results in clusters 4 and 6, we observed that the aggregation model converged faster with fewer aggregation groups. Cluster 6 has a lower number of aggregation groups, exhibiting a higher degree of model overfitting and instability compared to other clusters with the same number of vehicles. The aggregation model's convergence rate in cluster 6 was slower than that in 4, which aligns with actual traffic conditions. Additionally, as shown in Fig. 9, a smaller number of aggregation groups leads to more disturbances in the model. With a smaller sample size, the likelihood of the aggregated model being affected by

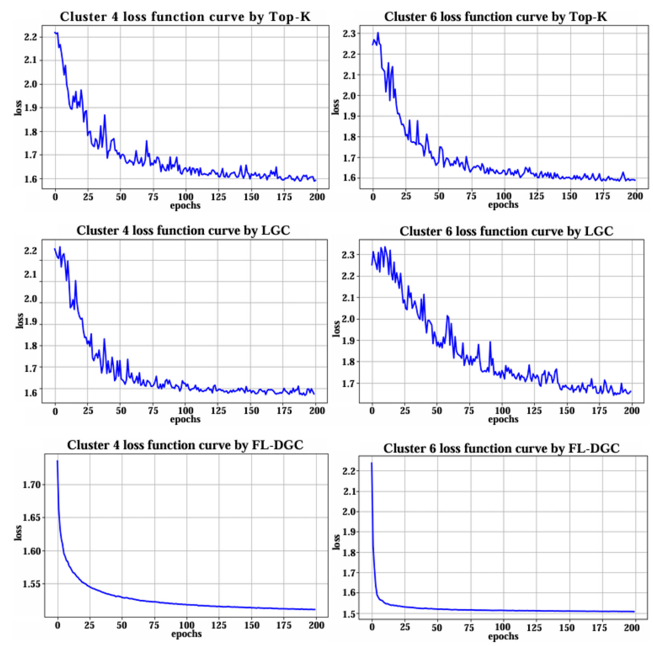

Fig. 9. Simulation comparison of clusters 4 and 6.

interference is high. Moreover, the results in Table VI indicate that an increase in the total amount of sensory data in clusters 4 and 6 results in more information sharing, improving the trained model's aggregate accuracy. Based on the simulation analysis, we discovered that the aggregated model's average accuracy in clusters 4 and 6 is similar, with a difference of 0.6%.

C. Group 3: Comparative Performance Evaluation

For these experiments, we used the MNIST dataset, unlike the experimental groups 1 and 2, which used CIFAR-10. In the preprocessing stage, we increased the model compression's sparsity to 99.9% [37] and compared FL-DGC to Top-K and LGC. By dividing the various clusters, we compared the aggregation model's fitting accuracy trends in complex and sparse regions. In addition, we selected different accuracy and gradient compression ratio benchmarks to demonstrate the proposed strategy's relative superiority. We conducted the experiments across different vehicle clusters, with the input data sample size increasing by 10% at each stage, allowing us to observe the model aggregation accuracy.

According to the gradient compression method, we followed the original settings provided in [37] and [47] for each specific training architecture, and the data volume gradient compression ratio is formulated as

$$CR = \frac{\text{size}\left(w_k^{\text{original}}\right)}{\text{size}\left(w_k^{\text{compression}}\right)} \tag{35}$$

where w_k^{original} and $w_k^{\text{compression}}$ represent the uncompressed and compressed gradient tensor sizes at the vehicle training node k, respectively.

Clusters 1 and 2 are densely populated with complex traffic scenarios. In these clusters, the number of clients in the vehicle

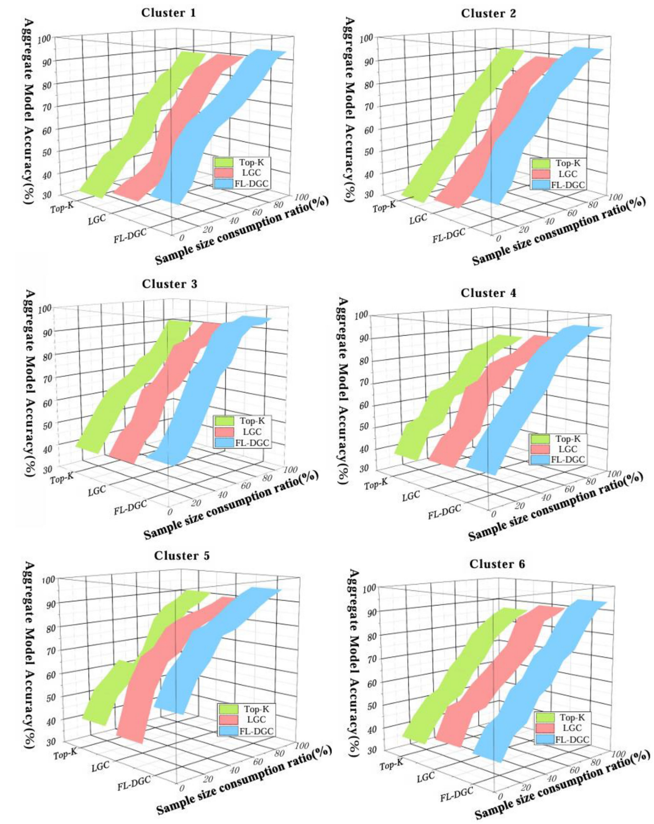

Fig. 10. Simulation comparison of the FL-DGC aggregation model's accuracy in different clusters.

is low, indicating that the network's dynamic topological changes are weak. Hence, we selected the model aggregation accuracy of 85% (±1%) as the baseline. In contrast, clusters 3 and 5 represent distinct urban environments, such as green parks and commercial districts. Despite having the same model training and aggregation quantities, these clusters have different vehicle numbers. Notably, the model converged faster in cluster 5, where the vehicle traffic is lower, than in cluster 3. Additionally, the proposed strategy exhibited a steeper convergence curve than the Top-K and LGC approaches, showcasing its enhanced performance.

Clusters 4 and 6 represent high-speed traffic and bridge scenarios. These clusters have the same number of vehicles but different model training and aggregation quantity settings. The qualitative analysis results indicate that the aggregation model in cluster 4 exhibited a higher convergence rate compared to 6 due to the influence of the number of aggregations.

Additionally, we compared various gradient compression methods from different literature. The experimental results for the gradient compression ratio in group 3 are presented in Table VII.

We examined the aggregated model's average accuracy trend using a gradient sparsity of 99.9% (i.e., the threshold for nonzero values). In Fig. 10, clusters 1 and 2 indicate that the Top-K method aggregates a lower number of models than the LGC local compression approach in complex traffic environments. This is due to poor learning accuracy during gradient descent caused by staleness, which is 9% and 9.9% lower than FL-DGC. We optimized (3) using dynamic gradient

TABLE VII
DISTRIBUTED TRAINING USING THE MNIST DATASET IN CLUSTER 4 AND 6

Batch size	Areas	Project	Accuracy	CR
128	Cluster 1	Baseline Top-K (Top-K Sampling) LGC (Local Gradient Compression) FL-DGC (Dynamic Gradient Compression)	85% (±1%) 83.7% -1.30% 85.4% +0.40% 93.6% +8.60%	1× 467× 538× 734×
256	Cluster 2	Baseline Top-K (Top-K Sampling) LGC (Local Gradient Compression) FL-DGC (Dynamic Gradient Compression)	85% (±1%) 84.1% -0.90% 87.7% +2.70% 94.7% +9.70%	1× 435× 561× 757×
128	Cluster 3	Baseline Top-K (Top-K Sampling) LGC (Local Gradient Compression) FL-DGC (Dynamic Gradient Compression)	80% (±1%) 79.40% -0.60% 81.70% +1.70% 90.20% +10.2%	1× 413× 459× 591×
256	Cluster 5	Baseline Top-K (Top-K Sampling) LGC (Local Gradient Compression) FL-DGC (Dynamic Gradient Compression)	80% (±1%) 82.80% +2.80% 83.20% +3.20% 92.40% +7.40%	1× 432× 509× 723×
128	Cluster 4	Baseline Top-K (Top-K Sampling) LGC (Local Gradient Compression) FL-DGC (Dynamic Gradient Compression)	80% (±1%) 76.6% -3.40% 78.8% -1.20% 91.8% +11.8%	1× 395× 422× 689×
256	Cluster 6	Baseline Top-K (Top-K Sampling) LGC (Local Gradient Compression) FL-DGC (Dynamic Gradient Compression)	80% (±1%) 73.7% -6.30% 81.9% +1.90% 92.1% +13.1%	1× 381× 493× 708×

compression factor correction. Based on the figure, we discovered that maintaining the model aggregation accuracy results in a CR of 734× and 757× with a different batch size (128\256). However, the Top-K and LGC compression ratios were slightly lower (i.e., 467× and 538×; 435× and 561×, respectively) in relatively complex traffic environments within clusters 1 and 2.

According to Fig. 10 and Table VII, we observed that the Top-K and LGC values were lower than FL-DGC (90.2%) when the sample size output was 50% with the 128 and 256 batches. We simulated increasing the complex environment's sample size and observed a stable fitting accuracy for the aggregation model. Notably, this can be observed in simple traffic scenarios, demonstrating that the aggregation model's accuracy was modified by the dynamic gradient compression factor. In addition, the FL-DGC method achieved a model compression of 178× and 132× higher than Top-K and LGC. Therefore, we conclude that FL-DGC can effectively improve the model aggregation accuracy in complex heterogeneous traffic environments. This is demonstrated by the sample size. In this instance, when the sample size reaches 60%, the aggregation model accuracy gradually stabilizes.

According to Fig. 10 and Table VII, we observed that after model training, the aggregation accuracy's stability and generalizability are maintained when the batch size is 128. Within the fast-paced and dynamic highway scenarios, the proposed strategy outperforms the Top-K and LGC compression ratios of 294× and 267× in cluster 4. When the batch size is 256, FL-DGC obtains a higher compression ratio than the 327× and 215× of the two quantization and sparsification methods.

Furthermore, based on the CR results in Table VII, we can assume that the gradient θ was calculated for each vehicle client n during FL in each training round. Hence, the Wasserstein distance used the parameters as the sparsification factors, the gradient compression factor δ_k was aggregated in (3), and the calculated parameter probability distribution was reduced by its similarity in (5). These results indicate that the FL-DGC strategy effectively maintains aggregation and gradient compression reliability in different scenarios and adaptive vehicle clusters. This demonstrates FL-DGC's superior compression ratios, stability, and generalizability.

V. CONCLUSION

Changes in network topology result in the limited and uneven distribution of resources between vehicles and base stations, leading to FL communication bottlenecks. Therefore, we proposed a dynamic gradient compression strategy based on the FL of adaptive vehicle clusters. While sending parameters to the base station from the vehicle's local client, the dynamic gradient compression factor was introduced as a sparsing threshold. As a result, the gradient, which participates in local aggregation, was efficiently selected, minimizing model learning errors and communication costs. During our study, we confirmed that our proposed strategy reduces FL communication overhead using six vehicle clusters, two datasets, and three sets of experiments, resulting in a global aggregation accuracy of 94.4% during redundant noise interference. In the future, we plan to investigate the dynamic relational network existing among the vehicles within vehicle clusters and the security aspects of the compression process.

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